Synopsis Report
On

InterviewRoom: An AI-Enabled Real-Time Technical Interview Platform

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Abstract

Technical interviews are a critical part of the hiring process in the software industry, yet conducting them effectively remains a challenge. Traditional interview methods rely on whiteboards, shared screens, or disconnected tools, which often fail to provide real-time collaboration, accurate code evaluation, and structured feedback. As a result, both interviewers and candidates face difficulties in assessing problem-solving skills, code quality, and overall performance in a realistic environment.

The proposed project aims to address these challenges by providing a unified and intelligent platform for conducting technical interviews. The system enables interviewers and candidates to collaborate in real time using a live coding editor, integrated communication tools, and shared problem statements. Artificial Intelligence is used to analyze the written code by evaluating factors such as time complexity, logical correctness, and coding standards, thereby offering meaningful insights beyond manual evaluation.

The platform also focuses on improving the post-interview process by generating automated feedback reports for both interviewers and candidates. These reports help interviewers make informed hiring decisions and assist candidates in understanding their strengths and areas for improvement. Additionally, the system maintains interview records, code snapshots, and performance metrics, making it useful for organizations to track and optimize their hiring process.

In summary, this project presents an AI-powered solution that simplifies technical interviews, enhances evaluation accuracy, reduces manual effort, and creates a more transparent and efficient hiring experience for all stakeholders.

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1. Introduction

1.1. Problem Statement:

Technical interviews play a crucial role in evaluating the problem-solving and coding skills of software engineering candidates. However, the current interview process often relies on fragmented tools such as shared screens, whiteboards, video conferencing platforms, and third-party coding websites. These tools lack seamless real-time collaboration, structured evaluation, and intelligent feedback, leading to an inefficient and inconsistent interview experience for both interviewers and candidates.

Interviewers frequently face challenges such as:

1. Lack of a unified platform that supports real-time collaborative coding and communication.
2. Difficulty in accurately evaluating code quality, time complexity, and logical correctness during live interviews.
3. Absence of automated and standardized feedback mechanisms after interviews.
4. Limited ability to track interview performance, code history, and candidate progress over time.
5. Inefficient handling of interview records, code snapshots, and evaluation metrics for hiring decisions.

These limitations result in increased manual effort, biased evaluations, and a less realistic assessment of a candidate's technical abilities. Therefore, there is a strong need for an AI-driven live coding interview platform that centralizes the interview process, enables real-time collaboration, and provides intelligent code analysis and feedback.

1.2. Objective:

The objectives of the InterviewRoom: An AI-Enabled Real-Time Technical Interview Platform are to:

1. Develop an intelligent platform for conducting real-time technical coding interviews.
2. Enable live collaborative coding between interviewers and candidates using an interactive code editor.
3. Provide real-time communication features such as chat and cursor sharing to simulate real interview environments.
4. Implement AI-based code analysis to evaluate correctness, time complexity, and coding standards.
5. Generate automated post-interview feedback reports for both interviewers and candidates.
6. Maintain interview records, code snapshots, and performance metrics for future reference.
7. Offer a responsive and user-friendly interface accessible across multiple devices.
8. Ensure secure authentication, data privacy, and role-based access control.
9. Design a scalable back-end system capable of handling multiple concurrent interviews efficiently.

1.3. Scope of Work

The InterviewRoom project covers the following scope of work:

1. User Registration and Authentication — Secure login and role-based access for interviewers and candidates.
2. Interview Room Creation — Setup of live coding interview rooms with predefined problems and configurations.
3. Real-Time Collaborative Coding — Live code editing, cursor sharing, and synchronized updates using real-time technologies.
4. AI Code Analysis — Automated evaluation of submitted code for logic, efficiency,

and best practices.

5. Communication Tools — Integrated chat functionality to support interviewer-candidate interaction.
6. Code Snapshot Management — Periodic saving of code states for review and analysis.
7. Feedback and Reporting — Generation and delivery of detailed post-interview feedback reports.
8. Data Storage — Storage of interview data, user profiles, and performance metrics using a NoSQL database.
9. Security and Scalability — Implementation of encryption, rate-limiting, and scalable architecture to support high user load.

2. Relevance to Computer Science and Engineering (CSE)

2.1. Artificial Intelligence and Machine Learning:
Artificial Intelligence is used to analyze source code written during interviews by evaluating logical correctness, time complexity, potential bugs, and coding standards. AI also helps in generating automated feedback reports, making the evaluation process more accurate and consistent.

2.2. Web Development and Full-Stack Engineering:
The project involves full-stack web development using modern front-end technologies (Next.js with React and TypeScript) and back-end frameworks (Node.js and Express). Real-time user interaction, state management, and responsive UI design demonstrate practical full-stack engineering skills.

2.3. Database Management Systems:
MongoDB is used to store interview room details, participant information, problem statements, code snapshots, and feedback data. This highlights the importance of efficient data modelling, storage, and retrieval in large-scale web applications.

2.4. Real-Time Systems and Distributed Computing:
Technologies such as WebSockets and Socket.io are used to enable real-time collaborative coding, live chat, and cursor sharing. Redis-based message handling supports scalability and low-latency communication, which is essential for distributed systems.

2.5. Software Engineering Principles:
The system follows key software engineering concepts such as modular architecture, separation of concerns, scalability, security, and maintainability. Practices like version control, API design, error handling, and performance optimization ensure the reliability of the platform.

3. Uniqueness of the Proposed System

The proposed InterviewRoom: An AI-Enabled Real-Time Technical Interview stands out by providing a realistic, intelligent, and collaborative technical interview experience that goes beyond traditional interview tools. Instead of relying on static code editors or screen sharing, the system integrates real-time collaboration with AI-based evaluation to closely simulate real-world technical interviews.

Key Unique Features:

• Live Collaborative Coding — Enables interviewers and candidates to write, edit, and view code simultaneously in a shared environment.

• AI-Based Code Analysis — Automatically evaluates code for correctness, time complexity, logical structure, and coding standards, reducing manual bias.

• All-in-One Interview Platform — Combines live coding editor, chat, problem statements, evaluation, and feedback in a single system.

• Real-Time Interaction — Supports live cursor sharing, chat messaging, and instant code synchronization for effective communication.

• Automated Feedback Generation — Provides detailed post-interview feedback reports to both interviewers and candidates.

• Code Snapshot Tracking — Periodically stores code states, allowing review of candidate progress throughout the interview.

• Interview Data Management — Maintains structured records of interviews, participants, problems, and evaluation results.

• Scalable Real-Time Architecture — Uses WebSocket and distributed systems to support multiple concurrent interviews with low latency.

• User-Friendly Experience — Clean and responsive interface designed for ease of use across desktops, tablets, and mobile devices.

• Practical over Generic — Unlike generic coding platforms, the system focuses on realistic interview scenarios, collaboration, and intelligent evaluation rather than simple code execution.

4. Technology Stack

The InterviewRoom: An AI-Enabled Real-Time Technical Interview Platform will be developed using the following technologies:

4.1. Front-End Technologies

• Next.js (React) with TypeScript — Component-based, server-rendered front-end for high performance, scalability, and type safety.
• tRPC with React Query — Type-safe client–server communication and efficient state management.
• Monaco Editor — Browser-based code editor providing an IDE-like experience for live coding interviews.
• Yjs (CRDT) — Enables real-time collaborative code editing and conflict-free synchronization.
• shadcn/ui & Tailwind CSS — For modern, responsive, and accessible user interface design.
• Responsive Design (CSS Grid / Flexbox) — Ensures seamless usability across desktops,

tablets, and mobile devices.

4.2. Back-End Technologies

• Node.js with Express (TypeScript) — Scalable and efficient server-side application and API development.
• Socket.io — Real-time bidirectional communication for live coding, chat, and cursor sharing.
• MongoDB — NoSQL database for storing interview rooms, participants, problems, code snapshots, and feedback data.
• Prisma ORM — Type-safe database access and schema management.
• Redis — Used for real-time message broadcasting and scalable Socket.io room management.
• JWT Authentication — Secure login and role-based session management for interviewers and candidates.

4.3. APIs (RESTful / Real-Time)

• Authentication API — Handles user registration, login, and role-based access control.
• Interview Room API — Creates and manages live interview rooms and participants.
• Real-Time Collaboration API — Supports live code synchronization, chat, and cursor updates.
• AI Code Analysis API — Analyses source code for correctness, complexity, and coding standards.
• Feedback & Reporting API — Generates and sends automated post-interview evaluation reports.
• Code Snapshot API — Stores and retrieves periodic code states for review and auditing.

4.4. Security

• Authentication & Authorization — Secure access using JWT and role-based permissions.
• HTTPS/SSL — Encrypted communication between client and server.
• API Rate Limiting — Protects the system from abuse and excessive requests.
• Data Validation & Sanitization — Ensures safe and reliable handling of user inputs.

5. Proposed System Architecture

The InterviewRoom : An AI-Enabled Real-Time Technical Interview follows a client–server architecture integrated with real-time communication services and AI-based code analysis. The system is divided into multiple layers to ensure modularity, scalability, and maintainability.

5.1. Front-End (Client Layer)

• Built using Next.js (React) with TypeScript to provide a responsive and interactive user interface.
• Provides users with functionalities such as interview room creation, live coding editor, chat interface, and interview dashboard.
• Integrates Monaco Editor for real-time coding and collaboration.
• Communicates with the back-end server using RESTful APIs and WebSocket over HTTPS.
• Ensures accessibility across devices including desktops, tablets, and mobile devices.

5.2. Back-End (Application Layer)

• Developed using Node.js and Express (TypeScript) to handle API requests, real-time communication, and business logic.
• Uses Socket.io for live code synchronization, chat messaging, and cursor sharing during interviews.
• Integrates AI services (OpenAI / Gemini API) to analyse code for correctness, efficiency,

and coding standards.
• Manages interview rooms, participant roles, authentication, and feedback generation.
• Designed to scale efficiently to support multiple concurrent interviews.

5.3. Database Layer

• Uses MongoDB as the primary NoSQL database.
• Stores interview room data, participant details, problem statements, code snapshots, and feedback reports.
• Supports efficient data retrieval, scalability, and high availability.

5.4. External Integrations

• AI APIs (OpenAI / Gemini API) — For automated code analysis and interview feedback generation.
• Email Service (Resend API) — For sending post-interview feedback reports to interviewers and candidates.
• Content Storage (Vercel Blob) — For storing problem statements and code snapshots.
• Redis — For real-time message broadcasting and scalable Socket.io communication.

5.5. Security Layer

• JWT Authentication — Secure user authentication and role-based access control.
• HTTPS/SSL — Encrypted client–server communication.
• Data Encryption — Protection of sensitive user and interview data.
• API Rate Limiting — Prevents misuse and ensures system stability.

6. How the Proposed System Stands Out:

6.1. Real-Time Collaborative Coding — Unlike traditional interview tools that rely on screen sharing or static editors, the system provides a live, synchronized coding environment where interviewers and candidates can collaborate in real time.
6.2. AI-Powered Code Evaluation — The platform uses Artificial Intelligence to analyse code for correctness, time complexity, potential bugs, and coding standards, offering deeper and more objective evaluation than manual reviews.
6.3. All-in-One Interview Platform — Combines live coding, chat, problem statements, code analysis, and feedback generation in a single platform, eliminating the need for multiple tools during interviews.
6.4. Automated Feedback Reports — Generates detailed post-interview feedback for both interviewers and candidates, improving transparency and reducing manual evaluation effort.
6.5. Interview Record and Code Snapshot Management — Automatically stores code snapshots and interview data, enabling review, auditing, and performance tracking over time.
6.6. Scalability and Real-Time Performance — Designed using WebSocket, Redis, and scalable back-end architecture to support multiple concurrent interviews with low latency.

7. Conclusion

The InterviewRoom: An AI-Enabled Real-Time Technical Interview Platform is designed to modernize and streamline the technical interview process. Traditionally, technical interviews depend on a combination of video conferencing tools, screen sharing, and third-party coding platforms, which often lack real-time collaboration, structured evaluation, and consistent feedback. This fragmented approach makes interviews inefficient and limits accurate assessment of a candidate's technical skills. The proposed system addresses these challenges by providing a centralized, intelligent platform for conducting live coding interviews.

By enabling real-time collaborative coding, integrated communication, and AI-based code analysis, the platform allows interviewers to evaluate candidates in a realistic and interactive environment. The system analyses code for correctness, efficiency, and coding standards, reducing manual effort and subjectivity in evaluation. Automated feedback generation further enhances transparency by providing detailed insights to both interviewers and candidates after the interview.

From a technical perspective, the project integrates multiple core areas of Computer Science and Engineering. The front-end offers a responsive and intuitive interface for live interviews, while the back-end manages real-time communication, interview sessions, and AI-driven analysis. A NoSQL database efficiently stores interview data, code snapshots, and feedback records, and external services support AI processing and automated reporting, ensuring reliability and scalability.

Academically, this project demonstrates the practical application of Artificial Intelligence, real-time systems, web development, database management, and software engineering principles to solve a real-world industry problem. Unlike traditional interview tools, InterviewRoom emphasizes collaboration, automation, and intelligent evaluation, making the technical hiring process more efficient, transparent, and effective.

8. References

[1] Hiware, Meher Surendra, et al. *REALCODEAI: An AI-Powered Real-Time Code Collaboration and Interview Platform*. International Journal of Scientific Research in Computer Science, Engineering and Information Technology, vol. 11, no. 6, 2025, pp. 49–54.

[2] Gomez, Nathalia, et al. *Virtual Interviewers, Real Results: Exploring AI-Driven Mock Technical Interviews on Student Readiness and Confidence*. arXiv, 2025.

[3] Sanjana, Silvia, et al. *A Systematic Review of AI-Driven Interviewing Systems for Technical and Professional Skills*. Proceedings of the ISCAP Conference, 2025, pp. 112–120.

[4] Sharma, Rakesh, et al. *Real-Time Dynamic Feedback in AI-Based Interview Systems*. Journal of Emerging Technologies and Innovative Research, vol. 12, no. 5, 2025, pp. 331–339.

[5] Nguyen, Truong Thanh Hung, et al. *SimInterview: LLM-Based Multilingual Interview Training System*. arXiv, 2025.

[6] Patil, Akshay, et al. *AI-Based Mock Interview System with NLP and Real-Time Feedback*. TIJER – International Research Journal, vol. 12, no. 5, 2025, pp. 215–221.

[7] Jadhav, S., D. Khan, and U. Waghmare. *AI-Driven Interview Coach for Personalized Interview Preparation*. IEEE International Conference on Intelligent Systems, 2025, pp. 402–408.

[8] Verma, Ankit, et al. *AI-Driven Real-Time Interview Simulation with Speech and Facial Analysis*. Indian Journal of Scientific and Technological Research, vol. 18, no. 25, 2025, pp. 2058–2066.

[9] Kulkarni, Rohit, et al. *Mock Interview Simulation Using Artificial Intelligence*. International Research Journal on Advanced Engineering Hub, vol. 3, no. 3, 2025, pp. 78–84.

[10] Mehta, Kunal, et al. *Effectiveness of AI-Driven Interview Preparation Tools*. Technical Studies in Intelligent Systems, vol. 7, no. 2, 2025, pp. 95–103.

[11] Gomez, Nathalia, et al. *Exploring AI-Driven Mock Technical Interviews on Student Preparedness*. ACM Digital Library Proceedings, 2025, pp. 188–196.

[12] Singh, Pawan, et al. *Mock Interviewer Powered by Artificial Intelligence*. Journal of International Informatics and Emerging Engineering, vol. 6, no. 4, 2025, pp. 144–150.

[13] Rao, Suresh, et al. *AI Tools for Interview Preparation: A Comparative Study*. International Journal of Artificial Intelligence Systems, vol. 9, no. 1, 2025, pp. 61–70.

[14] Choudhary, Neha, et al. *AI-Powered Real-Time Interview Simulator*. SkillCompiler Technical Journal, vol. 4, no. 2, 2025, pp. 29–36.

[15] Iyer, Mahesh, et al. *Collaborative Mock Interview Platforms with AI Evaluation*. Emerging Computer Science Technologies Review, vol. 10, no. 3, 2025, pp. 201–209.

[16] Banerjee, Anupam, et al. *Advances in AI-Enabled Technical Assessment Systems*. Computer Science Engineering Journal, vol. 14, no. 2, 2025, pp. 118–126.

[17] Thomas, Rebecca, et al. *AI-Driven Job Interview Systems with Personalized Feedback*. Education and Artificial Intelligence Journal, vol. 5, no. 1, 2025, pp. 41–49.

[18] Malhotra, Deepak, et al. *Real-Time Interview Analytics Using Artificial Intelligence*. Proceedings of the Computing and Interaction Systems Conference, 2025, pp. 302–309.

[19] Nair, Vishal, et al. *Technical Interview Preparation Trends Using AI*. International AI Systems Review, vol. 8, no. 4, 2025, pp. 177–185.

[20] Kapoor, Simran, et al. *Impact of AI-Based Interview Practice on Candidate Confidence*. Students and Artificial Intelligence Symposium Proceedings, 2025, pp. 66–73.